Native Language Use in English Learners' High School Science

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Introduction

Minnesota state law requires that teachers support students' multilingual abilities in addition to developing academic language appropriate for school (LEAPS Act, 2014).

Much has been done in educational research regarding multilingualism, largely focused on second language learning and literacy in K-12 education (Antón & DiCamilla,1999; Creese & Blackledge, 2010). Studies for English learning (EL) students learning science appear to focus more on science practices rather than issues of language transfer (Lee, Grapin & Haas, 2018).

This research works toward closing the gap between promoting multilingual education and teaching science to EL students.

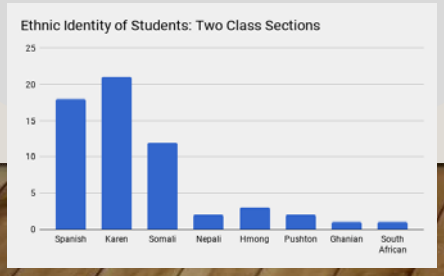
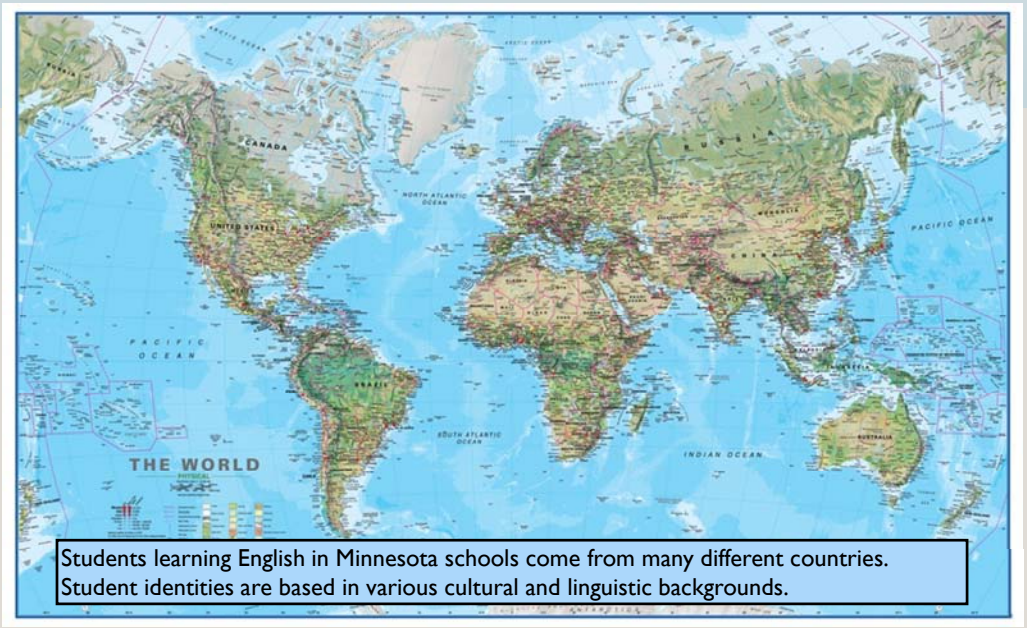


Figure 1. Student demographics – two sections of high school English Language and Physical Science class.



Methods

This study investigated the possibility of using an interactive video communication platform called Flipgrid to support EL students' native language use in science. The research question for this study was asking whether using a scaffolded oral assessment protocol would benefit students' language learning and science comprehension.

This co-taught course used curriculum designed as part of a larger study. Video data was collected as part of this larger teacher education project. This context was unique to the other participant sites in that it was the only co-taught class. Talking with the teachers, this large urban district was under scrutiny for underperformance and lack of support for Limited English Proficient (LEP) students. We discussed an alternative method of assessment using Flipgrid.

A quasi-experimental design was used to compare two sections of the English language and physical science course. The two sections of class were taught back-to-back. One class was prompted to record responses in both their home language and in English, The other section was not prompted for any particular language use.

Results

Student Flipgrid responses were scored based on a rubric for scientific understanding, and length of video was included as a measure of fluency. Analysis of paired t-tests showed no significant difference between the control group and the experimental group. Video data was reviewed to investigate what it was like for students and teachers in this co-taught class. Language learning was not addressed as often as science content. Different language groups were found to use home languages in different ways and at different times during classroom. There was not clear direction for what it meant to utilize both languages in the class.

Conclusions

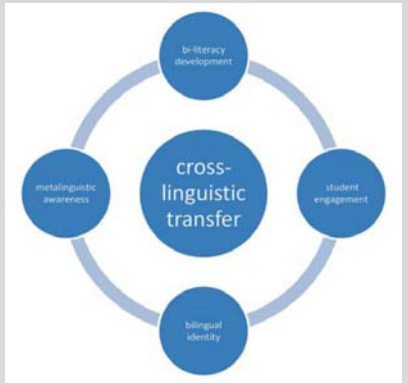


Figure 2. Elements of cross-linguistic transfer, based in bilingual education. "Conceptual framework for 50–50% bilingual pedagogy, diagramming the centrality of cross-linguistic transfer in relation to linguistic interdependence and bilingual learning" (Naqvi, Schmidt & Krickhan, 2014, p. 6).

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